11 FUNKSIYA LIMITI VA UZLUKSIZLIGI.

1. Funksiya limiti ta'rifidan foydalanib isbotlang: $\lim\limits_{x\to 3}(3x-5)=4$

Isbot. Ixtiyoriy $\varepsilon>0$ son uchun shungay $\delta>0$ topilib, $|x-3|<\delta$ tengsizlikni qanoatlantiruvchi barcha x lar uchun $|(3x-5)-4|<\varepsilon$ tengsizlik o'rinli bo'lishini ko'rsatishimiz kerak. Ixtiyoriy $\varepsilon>0$ son olaylik. $|(3x-5)-4|=|3x-9|=|3(x-3)|=3|x-3|<\varepsilon$ $\quad |x-3|<\frac{\varepsilon}{3}$. Agar $\delta<\frac{\varepsilon}{3}$ deb olsak, $|x-3|<\delta$ tengsizlikni qanoatlantiruvchi x lar uchun $|(3x-5)-4|<\varepsilon$ tengsizlik o'rinli bo'ladi.

Shu bilian $\lim\limits_{x\to 3}(3x-5)=4$ ekanligi isbotlandi. ▸

Funksiya limiti ta'rifidan foydalanib quyidagilarni isbotlang:

2. $\lim\limits_{x\to 1}(4x-1)=3$

3. $\lim\limits_{x\to \frac{\pi}{2}}\sin x=1$

4. $\frac{0}{0}$ ko'rinishidagi aniqmaslikni oching: $\lim\limits_{x\to 2}\frac{x-2}{x^2-3x+2}$

Yechish. $\lim\limits_{x\to 2}\frac{x-2}{x^2-3x+2}=\lim\limits_{x\to 2}\frac{x-2}{(x-2)(x-1)}=\lim\limits_{x\to 2}\frac{1}{x-1}=1$ ▸

? $\frac{0}{0}, \frac{\infty}{\infty}$ ko'rinishidagi aniqmasliklarni oching:

5. $\lim\limits_{x\to 3}\frac{x^2-9}{x^2-2x-3}$

6. $\lim\limits_{x\to \pi}\frac{tgx}{\sin 2x}$

7. $\lim\limits_{x\to 0}\frac{Sinx}{x}=1$ ajoyib limitdan foydalanib $\lim\limits_{x\to 0}\frac{Sin6x}{Sin7x}$ limitni toping:

Yechish. $\lim\limits_{x\to 0}\frac{Sin6x}{Sin7x}=\lim\limits_{x\to 0}\frac{\frac{Sin6x}{6x}6x}{\frac{Sin7x}{7x}7x}=\lim\limits_{x\to 0}\frac{6x}{7x}=\frac{6}{7}$ ▸

? $\lim\limits_{x\to 0}\frac{Sinx}{x}=1$ ajoyib limitdan foydalanib quyidagi limitlarni toping:

8. $\lim\limits_{x\to 0}\frac{Sin4x}{x}$

9. $\lim\limits_{x\to 0}\frac{1-Cos2x}{xSinx}$

10. $\lim\limits_{x\to \infty}\left(1-\frac{1}{x^2}\right)^x$ limitni hisoblang

Yechish. Bu limitni hisoblashda $\lim\limits_{x\to \infty}\left(1+\frac{1}{x}\right)^x=e$ ajoyib limitdan foydalanamiz:

$$\lim_{x\to \infty}\left(1-\frac{1}{x^2}\right)^x=\lim_{x\to \infty}\left[\left(1+\frac{1}{-x^2}\right)^{-x^2}\right]^{\frac{x}{-x^2}}=e^{\lim\limits_{x\to \infty}\left(\frac{-x}{x^2}\right)}=e^{-\lim\limits_{x\to \infty}\frac{1}{x}}=e^0=1.$$ ▸

? $\lim\limits_{x\to \infty}\left(1+\frac{1}{x}\right)^x=e$ ajoyib limitdan foydalanib limitlarni toping:

11. $\lim\limits_{x\to\infty}\left(1+\frac{5}{x}\right)^{2x}$

12. $\lim\limits_{x\to\infty}\left(\frac{2x+1}{2x-1}\right)^{x}$

13. $\lim\limits_{x\to\infty}\left(\frac{2x}{2x-3}\right)^{2-5x}$

14. $f(x)=\begin{cases} x+1, & agar\ \ 0<x<1 \\ 3x+1, & agar\ \ 1\le x\le 3 \end{cases}$ fuksiyaning $x=1$ nuqtada bir tomonli limitlarini toping:

Yechish. $f(1-0)=\lim\limits_{x\to 1-0} f(x)=\lim\limits_{x\to 1-0}(x+1)=2$

$f(1+0)=\lim\limits_{x\to 1+0} f(x)=\lim\limits_{x\to 1+0}(3x+1)=4$ ▸

Quyidagi fuksiyalarning ko‘rsatilgan nuqtalarida bir tomonli limitlarini toping:

15. $f(x)=\begin{cases} 3x, & agar\ \ -1\le x\le 1 \\ 2x, & agar\ \ 1<x\le 3 \end{cases}$ $x=1,\ x=2$ nuqtalarda

16. $y=\{x\},\quad \{x\}-x$ ning kasr qismi; $x=1,\ x=2,\ x=3$ nuqtalarda

17. $f(x)=x^2+x-2$ barcha $x\in(-\infty;+\infty)$ larda uzluksizligini ta’rifga binoan isbotlang.

Isbot. $f(x)=x^2+x-2$ barcha $x\in(-\infty;+\infty)$ larda

$$\lim_{\Delta x\to 0}(f(x+\Delta x)-f(x))=\lim_{\Delta x\to 0}((x+\Delta x)^2+(x+\Delta x)-2-(x^2+x-2))=$$

$$=\lim_{\Delta x\to 0}(2x\cdot\Delta x+\Delta x^2+\Delta x)=0$$

Demak, $f(x)$ barcha $x\in(-\infty;+\infty)$ larda uzluksiz ▸

Quyidagi funksiyalarning uzluksizligini ta'rifga binoan isbotlang:

18. $f(x) = \sin(3x+2)$, barcha $x \in (-\infty;+\infty)$ larda;

19. $f(x) = \dfrac{1}{x+1}$, barcha $(-1;+\infty)$ larda.

20. $f(x) = \begin{cases} x^2, & agar \quad -\infty < x \le 0 \\ (x-1)^2, & agar \quad 0 < x \le 2 \\ 5-x, & agar \quad 2 < x < \infty \end{cases}$ funksiyalarning uzilish nuqtalari va ularning turlarini aniqlang.

Yechish. $f(x)$ funksiya $(-\infty;0)$, $(0;2)$ va $(2;+\infty)$ intervallarda aniqlangan va uzluksiz bo'lgan elementar funksiyalar bilan berilgan. Demak, faqat $x_1 = 0$ *va* $x_2 = 2$ nuqtalarda uzulishga ega bo'lishi mumkin.

$x_1 = 0$ nuqta uchun chap va o'ng limitlarni hisoblaymiz:

$$\lim_{x\to 0-0} f(x) = \lim_{x\to 0-0} x^2 = 0;$$
$$\lim_{x\to 0+0} f(x) = \lim_{x\to 0+0} (x-1)^2 = 1; \quad f(0) = 0$$

Bu esa $x_1 = 0$ nuqtada $f(x)$ fuksiya o'ngdan birinchi tur uzilishga ega bo'lishini bildiradi. $x_2 = 2$ nuqta uchun:

$$\lim_{x\to 2-0} f(x) = \lim_{x\to 2-0} (x-1)^2 = 1$$
$$\lim_{x\to 2+0} f(x) = \lim_{x\to 2+0} (5-x) = 3 \quad f(2) = 1$$

bo'ladi.

$x_2 = 2$ nuqtada funksiya o'ngdan 1-tur uzilishga ega bo'ladi. ▸

Quyidagi funksiyalarning uzilish nuqtalari va ularning turlarini aniqlang. Grafiklarini yasang:

21. $y = \dfrac{4}{x-2}$

22. $f(x) = \begin{cases} \dfrac{x}{2}, & x \ne 2 \\ 0, & x = 2 \end{cases}$

23. $x^3+3x+1=0$ tenglama $[-1;\,0]$ kesmada yechimga ega ekanligini ko'rsating.

Yechish. $y=x^3+3x+1=0$ funksiya $[-1;\,0]$ da uzluksiz. Kesmaning uchlaridagi qiymatlari $f(-1)=-3$, $f(0)=1$ bo'lib, turli ishorali. Bolsano – Koshi teoremasiga binoan (-1; 0) da biror c nuqta topilib $f(x)=c^3+3c+1=0$ bo'lib, c berilgan tenglamalarning yechimi bo'ladi. ▸

Quyidagi tenglamalar ko'rsatilgan kesmalarda yechimga ega ekanligini ko'rsating:

24. $x^5-6x^2+3x-7=0;\quad [0;\,2]$

25. $3Sin^3x-5Sinx+1=0;\quad \left[0;\,\frac{\pi}{2}\right]$

26. $Cos^4x+3Cosx+1=0;\quad [0;\,\pi]$

27. $f(x)=Sinx\cdot Cos^2x-\sqrt{x+1}$ funksiya $[0;10]$ kesmada chegaralangan ekanligini isbotlang:

Isbot. $y=Sinx,\ y=Cos^2x\ \ va\ \ y=\sqrt{x+1}$ funksiyalarning har biri $[0;10]$ da uzluksiz bo'lganligi uchun, $f(x)=Sinx\cdot Cos^2x-\sqrt{x+1}$ funksiya ham [0;10] da uzluksiz. Shuning uchun Veyershtrass teoremasiga binoan $f(x)$ funksiya [0;10] da chegaralangan. ▸

Quyidagi funksiyalar ko'rsatilgan kesmalarda chegaralangan ekanligini isbotlang:

28. $f(x)=\dfrac{\sqrt{x^2+3x+1}}{x-1},\ [2;7]$

29. $f(x)=\sqrt{x^2+3x+1}\cdot Cos^7x,\ \ [0;2\pi]$

30. $\lim\limits_{\substack{x\to 0\\ y\to 3}} \frac{Sin(x^2\cdot y)}{x^2}$ funksiyaning limitini toping:

Yechish. $\lim\limits_{\substack{x\to 0\\ y\to 3}} \frac{\sin(x^2\cdot y)}{x^2} = \lim\limits_{\substack{x\to 0\\ y\to 3}} \frac{\sin(x^2\cdot y)}{x^2\cdot y}\cdot y = \lim\limits_{\substack{x\to 0\\ y\to 3}} y = 3$ ▸

Ko‘p o‘zgaruvchili funksiyalarning limitini toping:

31. $\lim\limits_{\substack{x\to -1\\ y\to 2}} \frac{1}{x^2+y^2}$

32. $\lim\limits_{\substack{x\to 3\\ y\to -5}} \frac{\sqrt{x}-y}{2x+y}$

33. $f(x,y)=\frac{1}{x^2+y^2}$ funksiyaning uzilish nuqtalarini toping.

Yechish. Funksiya $x=0$ va $y=0$ koordinatali nuqlalardan tashqari hanıma nuqtalarda aniqlangan va uzluksizdir. $O(0;0)$ koordinata boshi uzilish nuqtasi bo‘ladi. ▸

Quyidagi funksiyani $O(0;0)$ nuqtada uzluksizligini tekshiring:

34. $f(x,y)=\frac{xy}{\sqrt{xy+1}-1}$; $f(0,0)=2$

? $\frac{0}{0}, \frac{\infty}{\infty}$ ko'rinishidagi aniqmasliklarni oching:

35. $\lim\limits_{x\to\frac{\pi}{4}} \frac{\sin x - \cos x}{\cos 2x}$

36. $\lim\limits_{x\to 0} \frac{x}{\sqrt{1+3x}-1}$

37. $\lim\limits_{x\to 1} \frac{\sqrt[3]{x}-1}{\sqrt{x}-1}$

38. $\lim\limits_{x\to 0} \frac{\sqrt[3]{1+mx}-1}{x}$

39. $\lim\limits_{x\to 0} \frac{\sqrt{1+x}-\sqrt{1-x}}{x}$

40. $\lim\limits_{x\to \pi} \frac{\sqrt{1-tgx}-\sqrt{1+tgx}}{Sin2x}$

41. $\lim\limits_{x\to \infty} \frac{5x^3-7x}{1-2x^3}$

42. $\lim\limits_{x\to \infty} \frac{x^3-1}{x^2+1}$

43. $\lim\limits_{x\to \infty} \frac{\sqrt{x}-6x}{3x+1}$

44. $\lim\limits_{x\to -2} \frac{3x+6}{x^3+8}$

45. $\lim\limits_{x\to -1} \frac{x^2-x-2}{x^3+1}$

46. $\lim\limits_{x\to \pi+0} \frac{\sqrt{1+Cosx}}{Sinx}$

47. $\lim\limits_{x\to 7} \frac{2-\sqrt{x-3}}{x^2-49}$

48. $\lim\limits_{x\to 1} \frac{x^4-2x^3+2x^2-2x+1}{3x^4-5x^3+2x^2-x+1}$

49. $\lim\limits_{x\to 5} \frac{\sqrt{6-x}-1}{3-\sqrt{4+x}}$

50. $\lim\limits_{x\to 1} \frac{3x-2-\sqrt{4x^2-x-2}}{x^2-3x+2}$

51. $\lim\limits_{x\to 0} \frac{\sqrt{1+x+x^2}-\sqrt{1-x+x^2}}{x^2-x}$

? $\lim\limits_{x\to 0} \frac{Sinx}{x} = 1$ ajoyib limitdan foydalanib quyidagi limitlarni toping:

52. $\lim\limits_{h\to 0} \frac{Sin(x+h)-Sin(x-h)}{h}$

53. $\lim\limits_{x\to 0} \frac{1-Cosx}{x^2}$

54. $\lim\limits_{x\to 0} \frac{Sin4x}{\sqrt{x+1}-1}$

55. $\lim\limits_{x\to 0-0} \frac{\sqrt{1-Cos2x}}{x}$

56. $\lim\limits_{x\to 0}\dfrac{2x\cdot Sinx}{\sec x-1}$

57. $\lim\limits_{x\to 0}\dfrac{1-Cos2x+tg^2x}{xSinx}$

58. $\lim\limits_{x\to -2}\dfrac{\arcsin(x+2)}{x^2+2x}$

59. $\lim\limits_{x\to 0}\dfrac{1-Cos5x}{1-Cos3x}$

60. $\lim\limits_{x\to \frac{\pi}{2}}\dfrac{Cos5x}{Sin4x}$

61. $\lim\limits_{x\to 0}\dfrac{\sqrt{1+Cosx}-\sqrt{2Cosx}}{\sqrt{3+Cosx}-2\sqrt{Cosx}}$

? $\lim\limits_{x\to 0}\dfrac{Sinx}{x}=1$ ajoyib limitdan foydalanib quyidagi limitlarni toping:

62. $\lim\limits_{x\to \frac{\pi}{2}}\dfrac{1-Sinx}{(\frac{\pi}{2}-x)}$

63. $\lim\limits_{x\to \frac{\pi}{2}}\dfrac{Cos\frac{x}{2}-Sin\frac{x}{2}}{Cosx}$

? $\lim\limits_{x\to\infty}\left(1+\dfrac{1}{x}\right)^x=e$ ajoyib limitdan foydalanib limitlarni toping:

64. $\lim\limits_{x\to 0}(1-4x)^{\frac{1-x}{x}}$

65. $\lim\limits_{x\to\infty}\left(\dfrac{x+8}{x-2}\right)^x$

66. $\lim\limits_{x\to\infty}\left(\dfrac{x^2-2x+1}{x^2-4x+1}\right)^x$

67. $\lim\limits_{x\to\infty}\left(\dfrac{3x+1}{3x-1}\right)^{3x+1}$

19.

? Quyidagi fuksiyalarning ko‘rsatilgan nuqtalarida bir tomonli limitlarini toping:

68. $f(x)=\dfrac{3x+1}{x-1}$, $x=1$ nuqtada.

69. $f(x)=\begin{cases} x^2, & agar\ -1<x\le 2 \\ 2x+1, & agar\ 2<x<3 \end{cases}$ $x=2$ nuqtada

70. $f(x)=\dfrac{1}{x-2}$, $x=2$ nuqtada.

? Quyidagi funksiyalarning uzilish nuqtalari va ularning turlarini aniqlang. Grafiklarini yasang:

71. $y=1+\dfrac{|x+1|}{x+1}$

72. $y=2^{\frac{1}{x}}$

73. $f(x)=\begin{cases} 0.5x^2, & agar\ |x|<2 \\ 2.5, & agar\ |x|=2 \\ 3, & agar\ |x|>2 \end{cases}$

74. $f(x)=\begin{cases} Sinx, & x<0 \\ x, & 0\le x\le 2 \\ 0, & x>2 \end{cases}$

75. $f(x)=\begin{cases} -1, & x<0 \\ Cosx, & 0\le x\le \pi \\ 1-x, & x>\pi \end{cases}$

76. $f(x)=\begin{cases} -\dfrac{1}{x}, & x<0 \\ 1, & 0\le x<1 \\ x, & 1\le x\le 2 \\ 3, & 2<x\le 3 \end{cases}$

77. $f(x)=\begin{cases} -x, & x\le 1 \\ \dfrac{2}{x-1}, & x>1 \end{cases}$

78. $y=\dfrac{4}{4-x^2}$

79. $y=5^{\frac{4}{1-x}}+1$

80. $y=\dfrac{x+5}{x-3}$

81. $f(x)=9^{\frac{1}{2-x}}$

82. $y=\dfrac{x}{x+2}$

83. $y=2-\dfrac{|x|}{x}$

84. $f(x)=\begin{cases} 2, & agar\ x=0\ va\ x=\pm 2 \\ 4-x^2, & agar\ |x|<2 \\ 4, & agar\ |x|>2 \end{cases}$

85. $y=\dfrac{1}{1+2^{\frac{1}{x}}}$

86. $y=2^{\frac{1}{x-2}}$

? Ko‘p o‘zgaruvchili funksiyalarning limitini toping

79. $\lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{\ln(1-x^2-y^2)}{\sqrt{x^2+y^2}}$

80. $\lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{\sqrt{x^2+y^2+1}-1}{x^2+y^2}$

81. $\lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{Sin(x^3+y^3)}{x^2+y^2}$

82. $\lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{a-\sqrt{a^2-xy}}{xy}, \quad a\neq 0$

83. $\lim\limits_{\substack{x\to 3\\ y\to 0}} \dfrac{tg(xy)}{y}$

84. $\lim\limits_{\substack{x\to 0\\ y\to 0}} (x^2+y^2)\cdot Sin\dfrac{1}{xy}$

85. $\lim\limits_{\substack{x\to \infty\\ y\to a}} \left(1+\dfrac{y}{x}\right)^x$

86. $\lim\limits_{\substack{x\to 0\\ y\to 0}} \left(1+x^2+y^2\right)^{\frac{1}{x^2+y^2}}$

87. $\lim\limits_{x\to 0}\lim\limits_{y\to 0} \dfrac{x^3-y}{x^3+y}$

88. $\lim\limits_{y\to 0}\lim\limits_{x\to 0} \dfrac{y^2-x^2}{x^2+y^2}$